

first one

Dataset	titanic.csv
Date	2026-07-23 19:10
Selected Model	Random Forest
Dataset Health Score	96.8/100
Target Column	Survived
Task Type	Classification

Dataset Overview

Rows	891
Columns	11
Duplicate Rows	0
Total Missing Cells	0

AI Recommendations

- Remove 'PassengerId' — identifier column with near-unique values.
- Remove 'Name' — identifier column with near-unique values.
- High cardinality in 'Ticket' — use target/ordinal encoding.
- 'Age' has 19.87% missing — consider KNN imputation.
- Fill missing values in 'Embarked' using median/mode (0.22% missing).
- Label/Ordinal Encode 'Name' due to higher cardinality.
- One-Hot Encode 'Sex' (2 categories).
- Label/Ordinal Encode 'Ticket' due to higher cardinality.
- Label/Ordinal Encode 'Cabin' due to higher cardinality.
- One-Hot Encode 'Embarked' (3 categories).
- Apply StandardScaler or RobustScaler to numeric features.

Warnings

- 'Cabin' missing 77.1% values — consider dropping column.

Model Performance / Leaderboard

Model	Training_Time_sec	Accuracy	Precision	Recall	F1	ROC_AUC
Random Forest	2.241	0.8268	0.8282	0.8268	0.8242	0.9004
Logistic Regression	3.363	0.8156	0.8153	0.8156	0.8155	0.879
Decision Tree	0.04	0.7821	0.7826	0.7821	0.7823	0.7764
Naive Bayes	0.015	0.7821	0.7868	0.7821	0.7833	0.8551
SVM	0.521	0.6704	0.733	0.6704	0.6117	0.7246
KNN	0.043	0.6648	0.6602	0.6648	0.6496	0.6568

Best Model: Random Forest

— Model: Random Forest

— Training_Time_sec: 2.241

— Accuracy: 0.8268

— Precision: 0.8282

— Recall: 0.8268

— F1: 0.8242

— ROC_AUC: 0.9004

Final Conclusion

The dataset 'titanic.csv' contains 891 rows and 11 columns, with an overall health score of 96.8/100. Overall data quality is strong, with limited missing values, duplicates, or structural issues. For this classification task, 'Random Forest' was selected as the best-performing model on the held-out test set. Key metrics: Model: Random Forest, Training_Time_sec: 2.241, Accuracy: 0.8268, Precision: 0.8282, Recall: 0.8268, F1: 0.8242, ROC_AUC: 0.9004. Limitations: results are based on a single train/test split and default hyperparameters unless tuning was applied; performance may vary with more data, feature engineering, or cross-validation. For production deployment, monitor for data drift and retrain periodically.